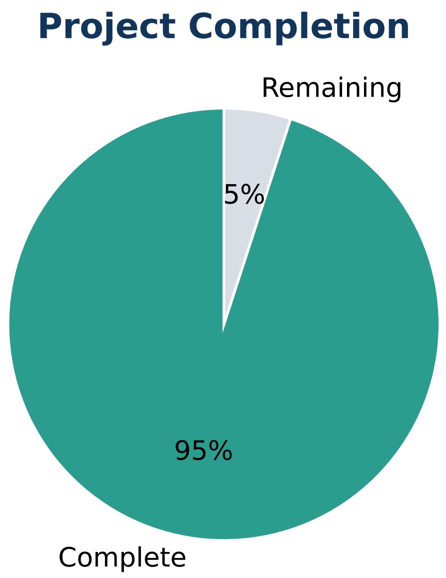
SOC-Ready Explainable IDS Triage

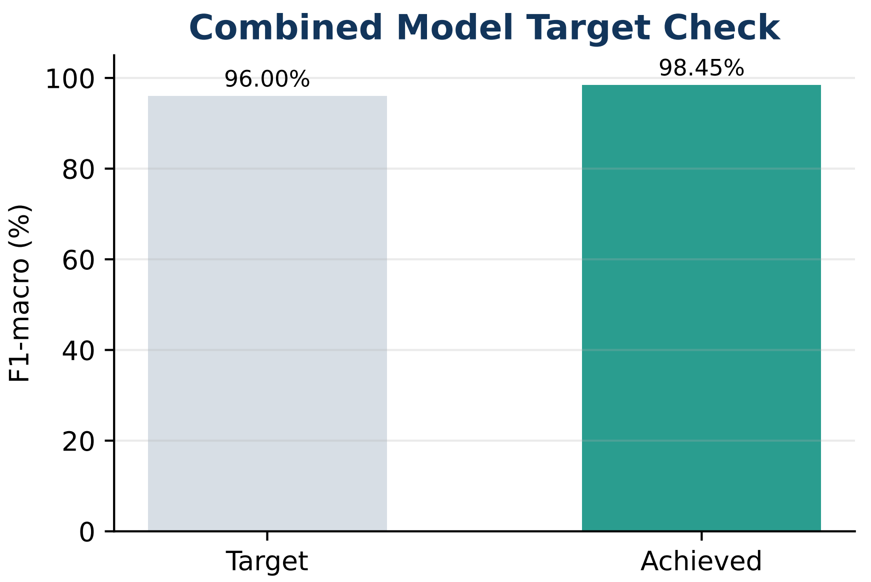
Professor Results Summary

Executive Status

The project implementation is functionally complete. It supports CICIDS2017, BoT-IoT, and a unified combined mode. The combined model is the primary dissertation result because it trains one IDS model on both datasets simultaneously.

Overall completion	Combined F1-macro	Combined accuracy	Test coverage
95%	**98.45%**	**99.88%**	**96.09%**
Remaining: live Wazuh evidence	Target > 96% achieved	Held-out combined split	48 tests passed





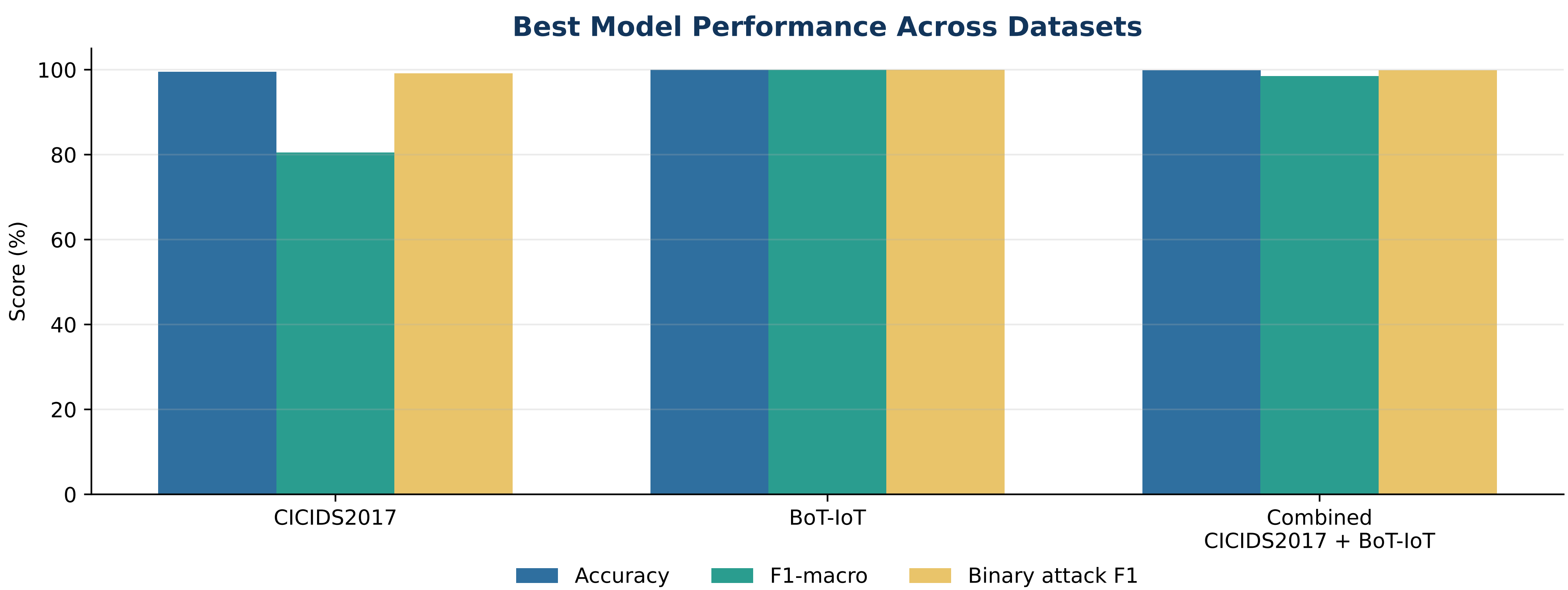
Remaining operational step

Deploy Wazuh/OpenSearch on an authorised Linux VM/server and capture the dashboard/evidence screenshots.

Main Achieved Results

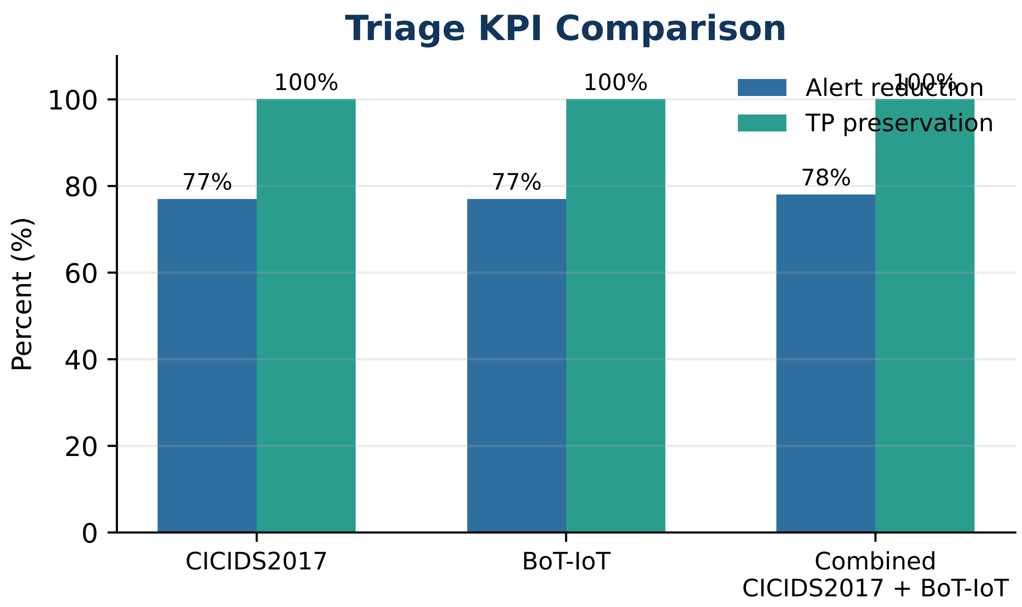
Best model performance by dataset

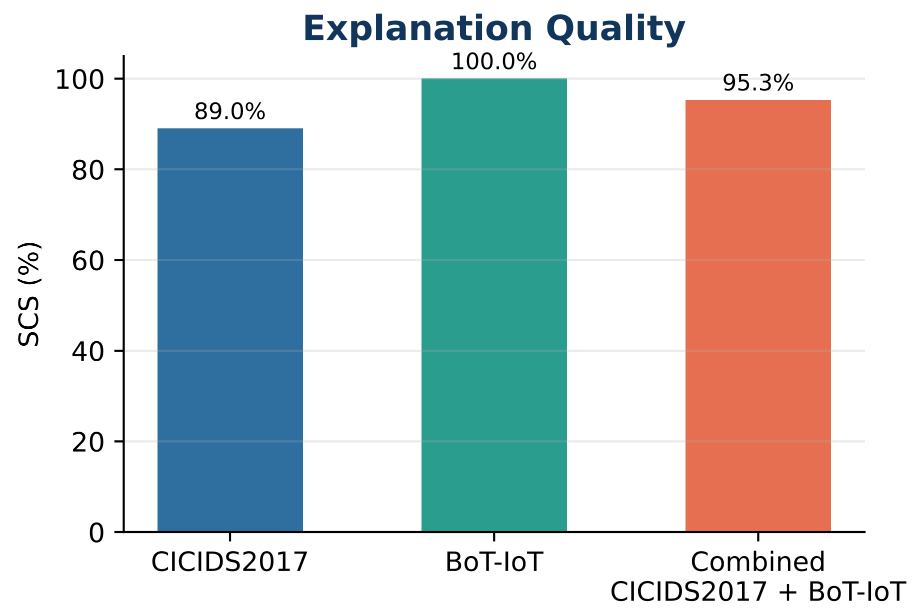
Dataset	Best model	Accuracy	F1-macro	Binary attack F1	Alert reduction	TP preservation	Explanation SCS
CICIDS2017	Random Forest	99.50%	80.47%	99.16%	77.0%	100.0%	89.00%
BoT-IoT	Random Forest	100.00%	100.00%	100.00%	77.0%	100.0%	100.00%
Combined CICIDS2017 + BoT-IoT	**XGBoost**	99.88%	**98.45%**	99.89%	78.0%	100.0%	95.30%

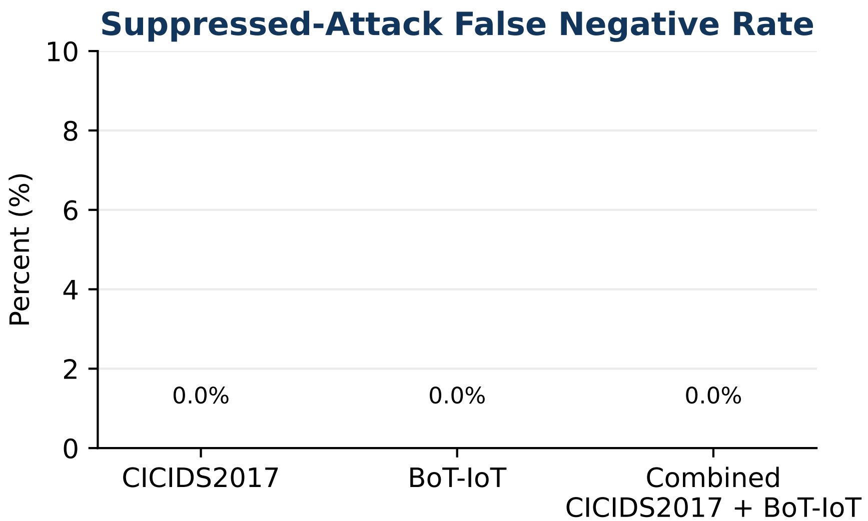


Triage and Explainability Outcomes

Alert-fatigue reduction with safety preservation







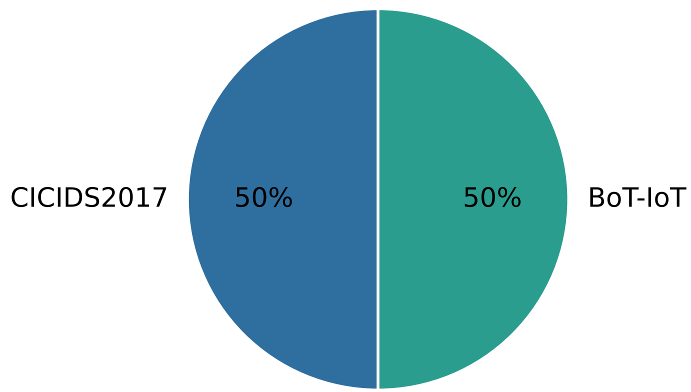
Interpretation

- Alert reduction means fewer repeated or low-value alerts reach the analyst.
- TP preservation at 100% means real attack evidence remains represented after triage.
- Combined explanation SCS is 95.3%, indicating strong completeness, actionability, and conciseness.

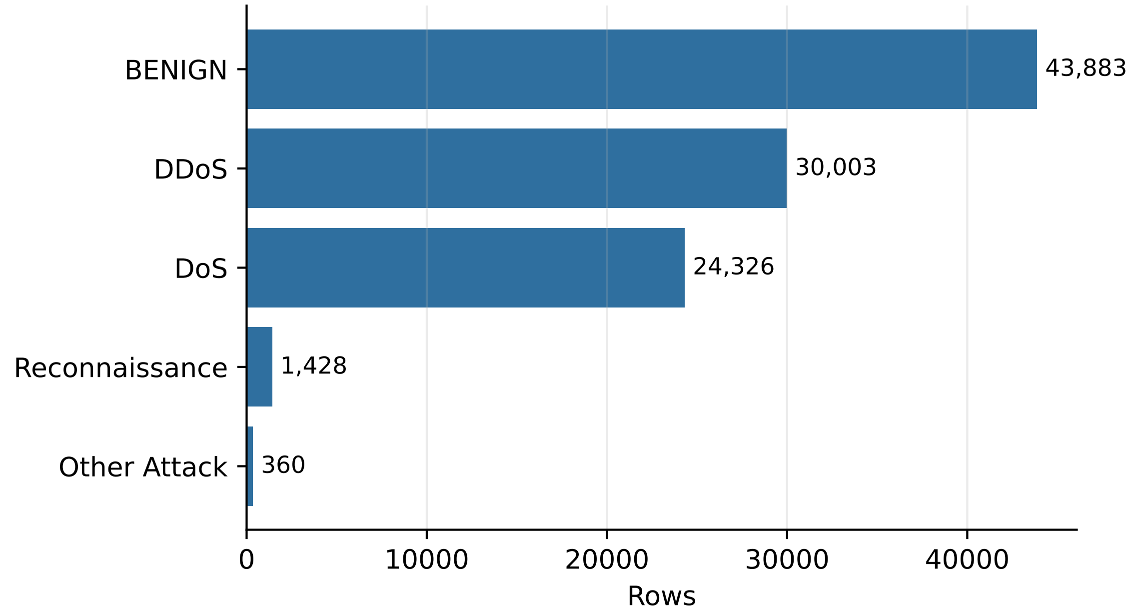
Combined Dataset Composition

Unified model input distribution

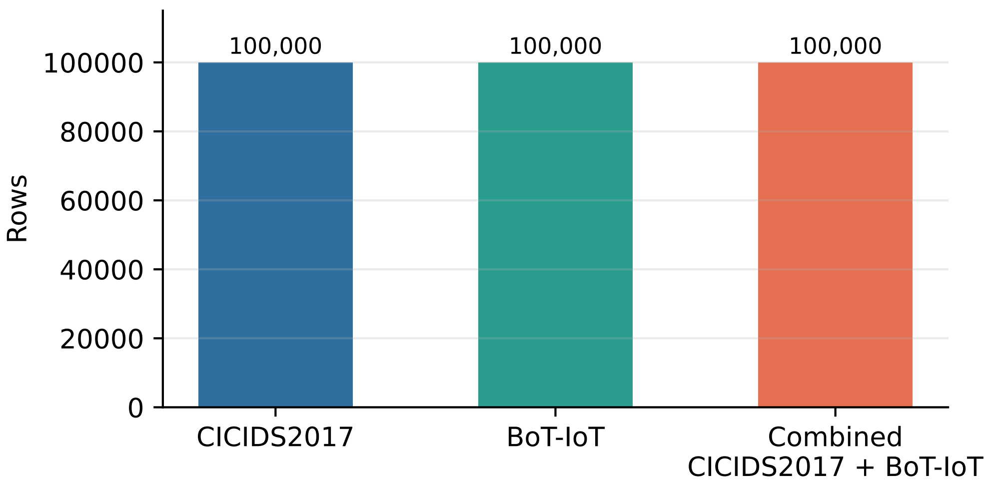
Source Dataset Split



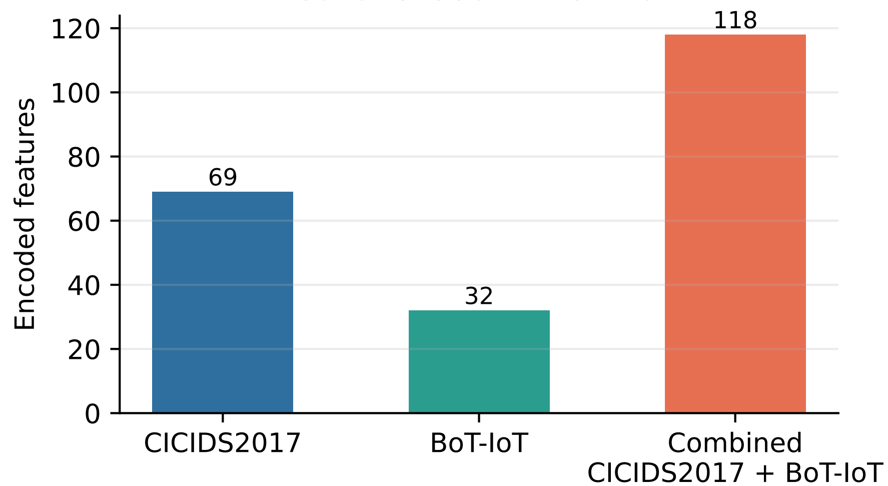
Combined Class Distribution



Retained Rows Per Run

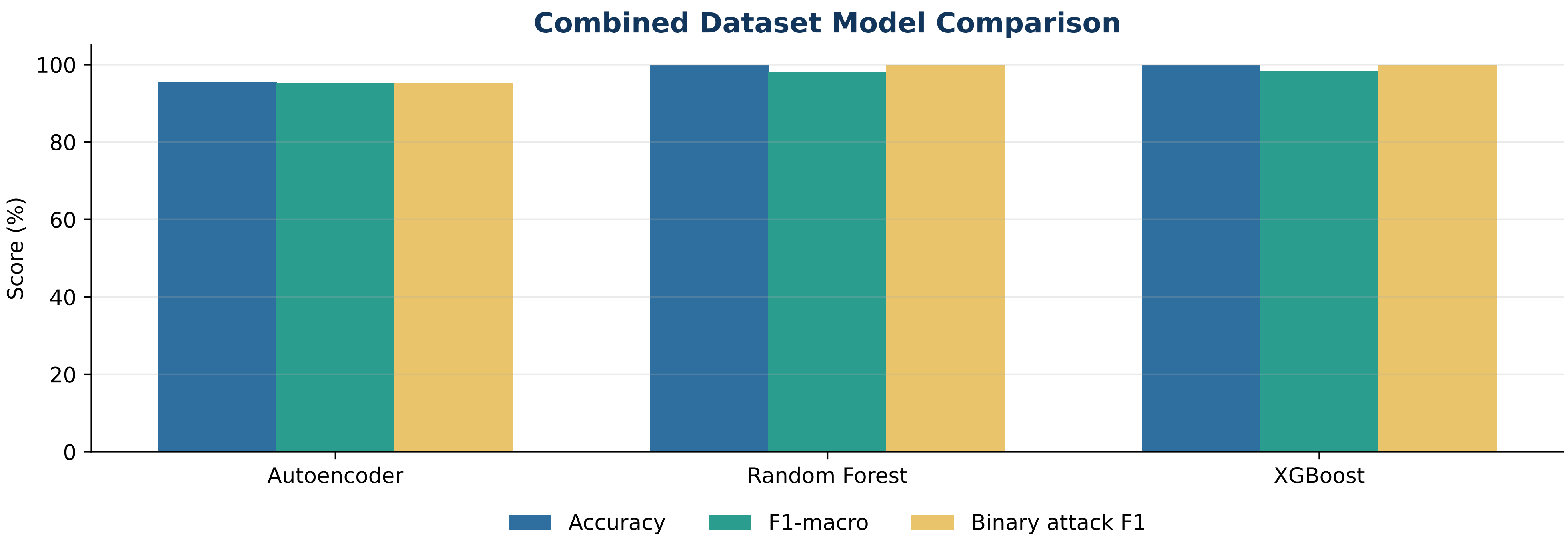


Feature Count Per Run



Combined Model Selection

Why XGBoost is the selected downstream artifact



Final Project Outcome

- The simultaneous two-dataset requirement is met: combined F1-macro is 98.45%, above the 96% target.
- The triage layer reduces alert volume by 78% while preserving 100% of represented true-positive attack evidence.
- Local Wazuh validation passes; only live Wazuh/OpenSearch deployment and screenshots remain.

Important Reporting Note

CICIDS2017 15-class macro-F1 is lower because rare attack classes are difficult under the configured 100,000-row sample. However, binary attack detection remains high at 99.16%, and the primary unified combined model reaches 98.45% macro-F1.